

8.10

$$M_{X_i}(t) = 1/2e^0 + 1/3e^t + 1/6e^{2t} = \frac{1}{2} + \frac{1}{3}e^t + \frac{1}{6}e^{2t}$$

$$M_S(t) = M_{X_1}(t)M_{X_2}(t) = (\frac{1}{2} + \frac{1}{3}e^t + \frac{1}{6}e^{2t})^2$$

$$\rightarrow M_S(t) = \frac{1}{4} + \frac{1}{6}e^t + \frac{1}{12}e^{2t} + \frac{1}{6}e^t + \frac{1}{9}e^{2t} + \frac{1}{18}e^{3t} + \frac{1}{12}e^{2t} + \frac{1}{18}e^{3t} + \frac{1}{36}e^{4t}$$

$$\rightarrow M_S(t) = \frac{1}{4} + \frac{1}{3}e^t + \frac{1}{6}e^{2t} + \frac{1}{9}e^{2t} + \frac{1}{9}e^{3t} + \frac{1}{36}e^{4t}$$

$$\rightarrow M_S(t) = \frac{1}{4} + \frac{1}{3}e^t + \frac{5}{18}e^{2t} + \frac{1}{9}e^{3t} + \frac{1}{36}e^{4t}$$

$$P(S=0) = \frac{1}{4}, P(S=1) = \frac{1}{3}, P(S=2) = \frac{5}{18}, P(S=3) = \frac{1}{9}, P(S=4) = \frac{1}{36}$$

8.15

$$\text{Area of trapezoid } A = \frac{\text{top} + \text{bottom}}{2} = \frac{3}{2}$$

$$f_{X,Y}(x,y) = \frac{2}{3}, (x,y) \in D, 0 \text{ otherwise.}$$

We have the bounds $0 \leq x \leq 1, 0 \leq y \leq 1$ and $1 \leq x \leq 2, 0 \leq y \leq 2-x$

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$E(XY) = \int_0^1 \int_0^1 \frac{2}{3}xydx dy + \int_1^2 \int_0^{2-x} \frac{2}{3}xydy dx$$

$$E(XY) = \int_0^1 \frac{1}{3}ydy + \int_1^2 \frac{1}{3}x(2-x)^2dx$$

$$E(XY) = \int_0^1 \frac{1}{3}ydy + \int_1^2 \frac{1}{3}(4x - 4x^2 + x^3)dx$$

$$E(XY) = \frac{1}{6} + \frac{1}{3}(2x^2 - \frac{4}{3}x^3 + \frac{1}{4}x^4)|_1^2$$

$$E(XY) = \frac{1}{6} + \frac{1}{3}(8 - \frac{32}{3} + 4 - (2 - \frac{4}{3} + \frac{1}{4}))$$

$$E(XY) = \frac{1}{6} + \frac{1}{3}(10 - \frac{28}{3} - \frac{1}{4})$$

$$E(XY) = \frac{1}{6} + \frac{1}{3}(\frac{5}{12})$$

$$E(XY) = \frac{11}{36}$$

$$E(X) = \int_0^1 \int_0^1 \frac{2}{3}xdydx + \int_1^2 \int_0^{2-x} \frac{2}{3}xdydx$$

$$E(X) = \int_0^1 \frac{2}{3}xdx + \int_1^2 \frac{2}{3}2x - x^2dx$$

$$E(X) = \frac{1}{3} + \frac{2}{3}(x^2 - \frac{1}{3}x^3)|_1^2$$

$$E(X) = \frac{1}{3} + \frac{2}{3}(4 - \frac{8}{3} - (\frac{2}{3}))$$

$$E(X) = \frac{1}{3} + \frac{2}{3}(4 - \frac{10}{3})$$

$$E(X) = \frac{1}{3} + \frac{4}{9}$$

$$E(X) = \frac{7}{9}$$

$$E(Y) = \int_0^1 \int_0^1 \frac{2}{3} y dx dy + \int_1^2 \int_0^{2-x} \frac{2}{3} y dy dx$$

$$E(Y) = \frac{1}{3} + \int_1^2 \frac{1}{3} (4 - 4x + x^2) dx$$

$$E(Y) = \frac{1}{3} + \frac{1}{3} (4x - 2x^2 + \frac{1}{3}x^3)|_1^2$$

$$E(Y) = \frac{1}{3} + \frac{1}{3} (\frac{8}{3} - (2 + \frac{1}{3}))$$

$$E(Y) = \frac{1}{3} + \frac{1}{9}$$

$$E(Y) = \frac{4}{9}$$

$$\rightarrow Cov(X, Y) = \frac{11}{36} - \frac{7}{9} \frac{4}{9} = \frac{11}{36} - \frac{28}{81} = -\frac{13}{324}$$

Based on the description of this experiment the Covariance should be negative because as X increases, Y tends to decrease (specifically for $x > 1$).

8.26

a

$$P(X = G) = \frac{4}{9}$$

$$P(X = Y) = \frac{3}{9}$$

$$P(G \text{ then } Y) = \frac{4}{9} \frac{3}{9} = \frac{12}{81} = \frac{4}{27}$$

With n draws, there are $n-1$ consecutive pairs. $E(X_n) = (n-1) * P(G \text{ then } Y) = (n-1) \frac{4}{27}$

b

We can model this as a geometric distribution.

Let NW denote the number of draws before we see a white.

$$P(NW = k) = (1 - \frac{2}{9})^{k-1} \frac{2}{9}$$

$$E(NW) = \frac{1-p}{p} = \frac{7}{2} \text{ for } k \text{ failures.}$$

Then the probability a non white is green is $\frac{4}{7}$

$$\text{Therefore } E(Y) = E(NW) \frac{4}{7} = 2$$

An intuitive approach is there are twice as many greens as there are white, meaning that for each white there are 2 greens.

8.37

a

Let $T_i \geq 1$ denote that there is at least 1 copy of the i th toy.

$$P(T_i \geq 1) = 1 - P(T_i = 0) = 1 - (\frac{9}{10} \frac{8}{9})^4 = 1 - \frac{4^4}{5^4} = 1 - \frac{256}{625} = \frac{369}{625}$$

Here $P(T_i \geq 1) = P(J_i = 1) = E(J_i)$ where J_i is the indicator variable for the i th toy.

$$E(X) = \sum_{i=1}^{10} E(J_i) = \frac{3690}{625} = \frac{738}{125}$$

b

$$Var(X) = \sum Var(J_i) + \sum_{i=1}^{10} \sum_{k \neq i} Cov(J_i, J_k)$$

$$\rightarrow Var(X) = 10Var(J_1) + 10 * 9Cov(J_1, J_2)$$

$$\text{Given in the textbook, } Var(J_1) = p(1-p) = \frac{256}{625} \frac{369}{625} = \frac{94464}{390625}$$

$$\rightarrow Var(X) = 10(\frac{94464}{390625}) + 10 * 9Cov(J_1, J_2)$$

$$Cov(J_1, J_2) = E(J_1 J_2) - E(J_1)E(J_2)$$

$$E(J_1 J_2) = 1 - (P(J_1 = 0) + P(J_2 = 0) - P(J_1 = 0 \wedge P(J_2 = 0)))$$

$$E(J_1 J_2) = 1 - (\frac{256}{625} + \frac{256}{625} - (\frac{8}{10} \frac{7}{9})^4)$$

$$E(J_1 J_2) = 1 - (\frac{512}{625} - (\frac{28}{45})^4)$$

$$E(J_1 J_2) = \frac{1356049}{4100625}$$

$$Cov(J_1, J_2) = \frac{1356049}{4100625} - \frac{369^2}{625}$$

$$Cov(J_1, J_2) = \frac{1356049}{4100625} - \frac{136161}{390625}$$

$$Cov(J_1, J_2) = \frac{-45821696}{2562890625}$$

$$\rightarrow Var(X) = \frac{944640}{390625} + 90\left(\frac{-45821696}{2562890625}\right)$$

$$\rightarrow Var(X) = \frac{944640}{390625} + \frac{-91643392}{56953125}$$

$$\rightarrow Var(X) = \frac{9217024}{11390625} \approx 0.8091763182441701$$

9.6

$$15min = 900sec$$

$$P(S_{64} \leq 900) = P(Z \leq \frac{900-15*64}{4\sqrt{64}})$$

$$P(S_{64} \leq 900) = P(Z \leq \frac{-60}{32})$$

$$P(S_{64} \leq 900) = P(Z \leq \frac{-15}{8}) = \Phi(-1.875) \approx \Phi(-1.88) = 0.0301$$

9.21

$$E(X_i) = E(Y_i) = 2$$

$$Var(X_i) = 2, Var(Y_i) = 3$$

$$P(\sum_{i=1}^{500} X_i > \sum_{i=1}^{500} Y_i + 50)$$

$$\text{Let } S_n = \sum X_i - \sum Y_i$$

$$P(\sum_{i=1}^{500} X_i > \sum_{i=1}^{500} Y_i + 50) = P(S > 50)$$

$$\rightarrow P\left(\frac{S}{\sigma\sqrt{n}} > \frac{50}{\sigma\sqrt{n}}\right)$$

$$n = 500, \sigma^2 = Var(X_i) + Var(Y_i) = 5$$

$$\sigma\sqrt{n} = 50$$

$$\rightarrow P\left(\frac{S}{50} > 50\right) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$